



Spin ground state and optical excitations in the vacancy/oxygen defect in the hexagonal BN

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Abstract

We investigate the V_{BO_N} defect in 2D hexagonal BN, which consists of a B-vacancy and an O-substituted N-atom, as a potential spin qubit for quantum computing. Our calculations confirm the existence of a triplet state in this defect, as previously reported. However, we find that the triplet state is dynamically unstable due to imaginary frequency modes in the phonon spectrum. By distorting the structure based on one of these modes, we induce a transition to a stable singlet state with lower total energy. Our Bethe-Salpeter calculations reveal that the singlet state exhibits a sharp optical excitation peak at 2.5 eV, making it promising for single-photon emission applications.

Introduction

2D hexagonal boron nitride (hBN) is a wide-band-gap semiconductor considered a promising host for defects with single-photon-emitter (SPE) and spin-qubit (SQ) functionalities, both requiring local electronic states in the band gap, with SQs additionally needing a non-zero (preferably triplet) spin state. This work focuses on the VBON defect, formed by a B-vacancy and an O-substituted neighboring N atom. The defect features two N atoms with dangling bonds, favorable for spin polarization, and an even number of valence electrons that support triplet formation. Previous first-principles calculations [1] identified a triplet state but suggested the defect does not create local electronic states in the band gap, limiting its potential as a SPE. We reevaluate the V_{BO_N} defect using density functional theory (DFT) for energy and phonon spectra, the GW method for an accurate electronic structure, and the Bethe-Salpeter equation (BSE) to calculate the optical excitation spectrum.

Triplet or Singlet?

We model the defected hBN 5x5x1 supercell by removing a B atom and replacing N with O. DFT calculations show the system adopts a triplet state, with an electronic structure similar to Ref. 1.

Phonon analysis of the triplet VBON reveals imaginary modes, indicating dynamical instability. Distorting the structure according to these modes leads to a singlet-state VBON, which has 0.24 eV lower energy than the triplet. The singlet VBON is stable, with no imaginary phonon modes.

The singlet VBON formation energy is 2.33 eV under both N-rich and N-poor conditions, much lower than the VBCN defect (9.234 eV), suggesting higher thermodynamic stability for V_{BO_N} .

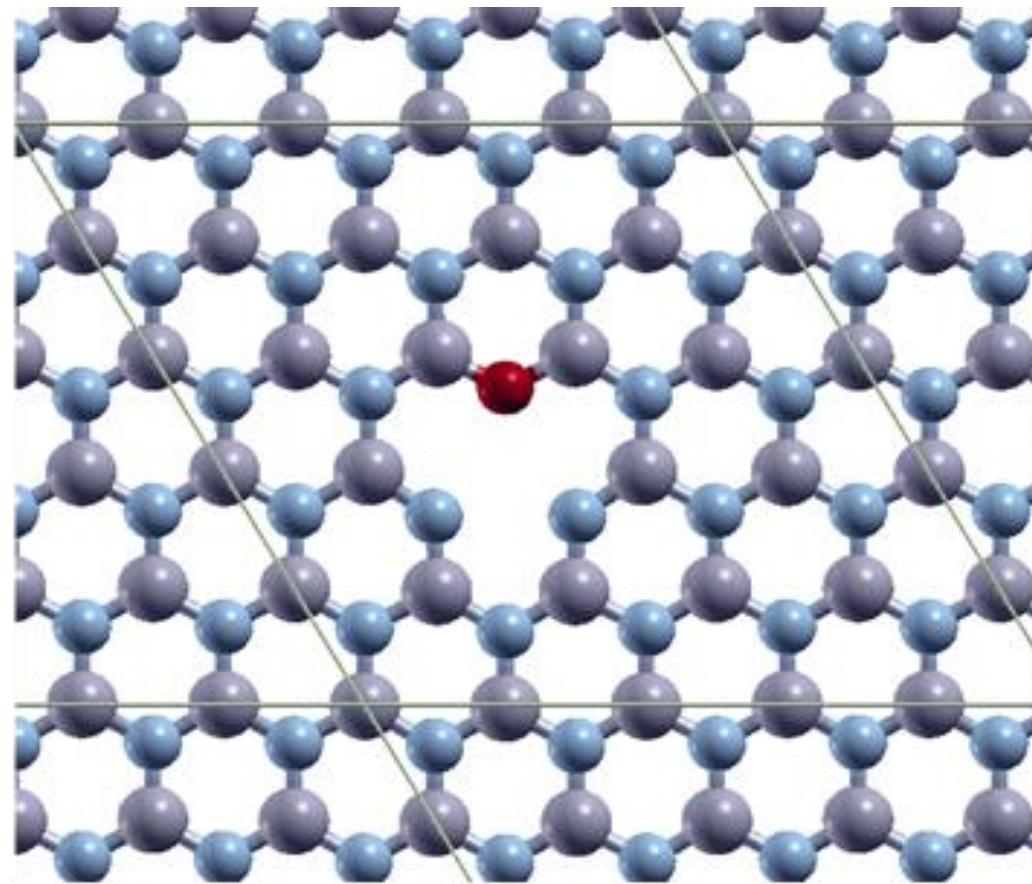


Fig. 1. The V_{BO_N} defect in hBN. The grey, blue, and red balls represent the B, N, and O atoms, respectively. Light green lines indicate the supercell boundaries.

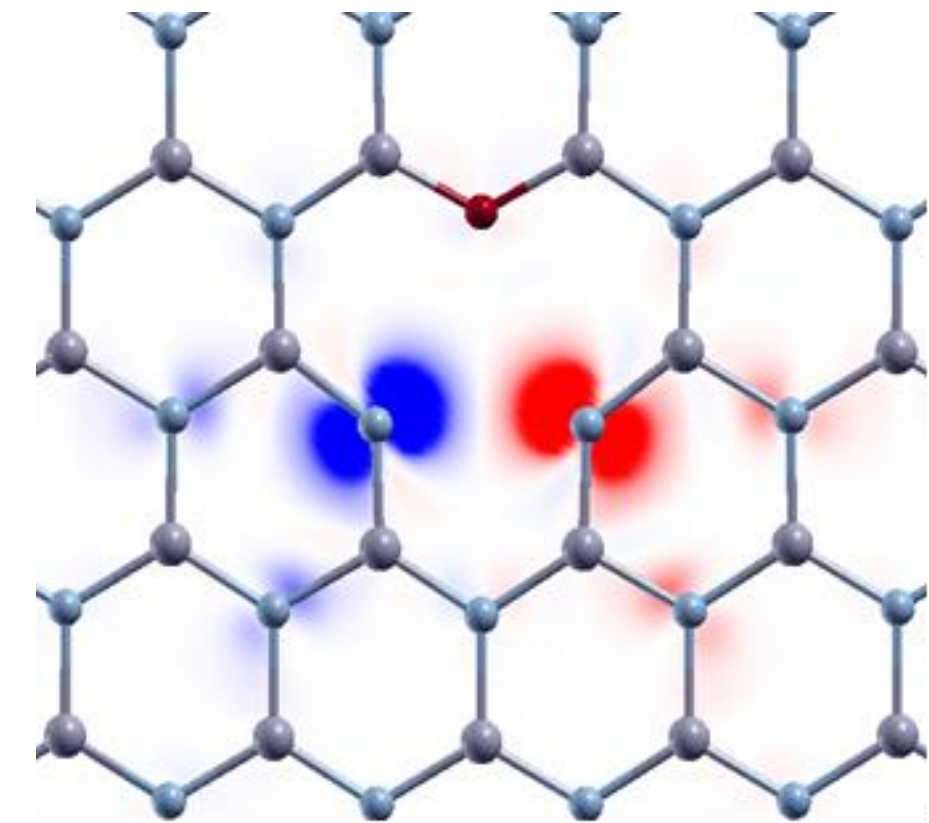


Fig. 2. In-plane cut of the spin density $\rho_s(r)$ in the singlet state of the V_{BO_N} defect. The blue and red areas correspond for the spin-up and spin-down states, respectively.

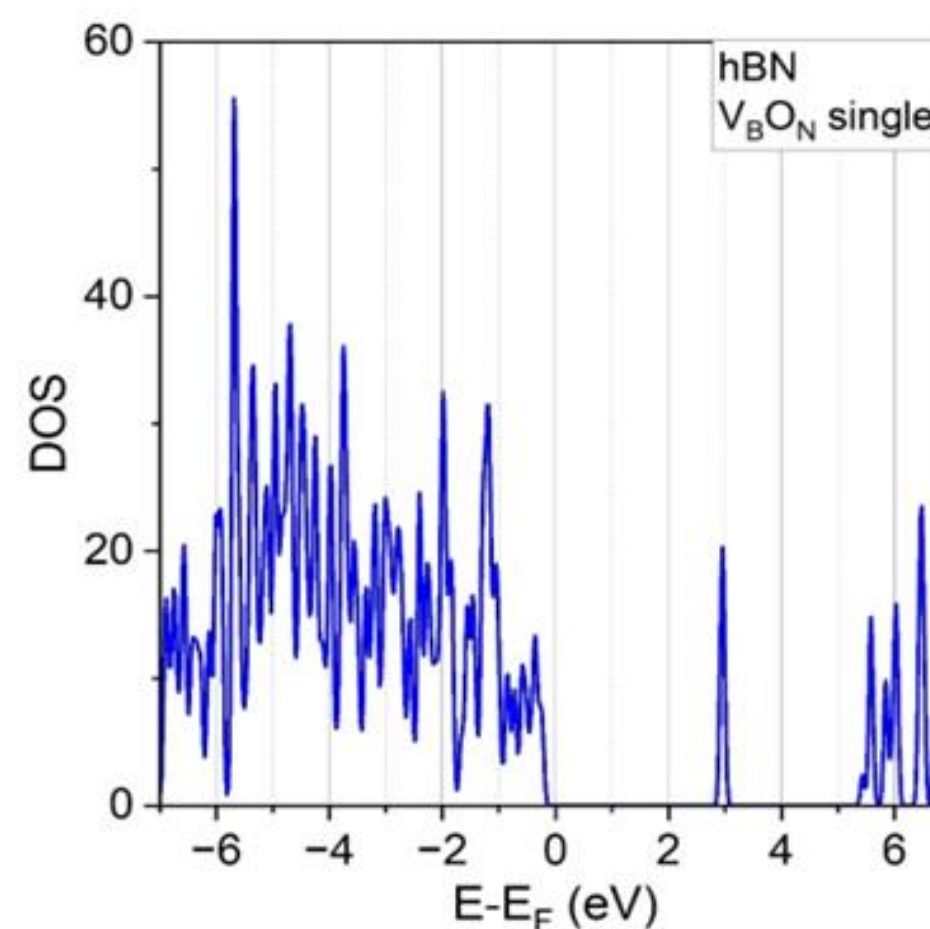


Fig. 3. Densities of the electronic states calculated for the singlet state of the V_{BO_N} defect using the GW method.

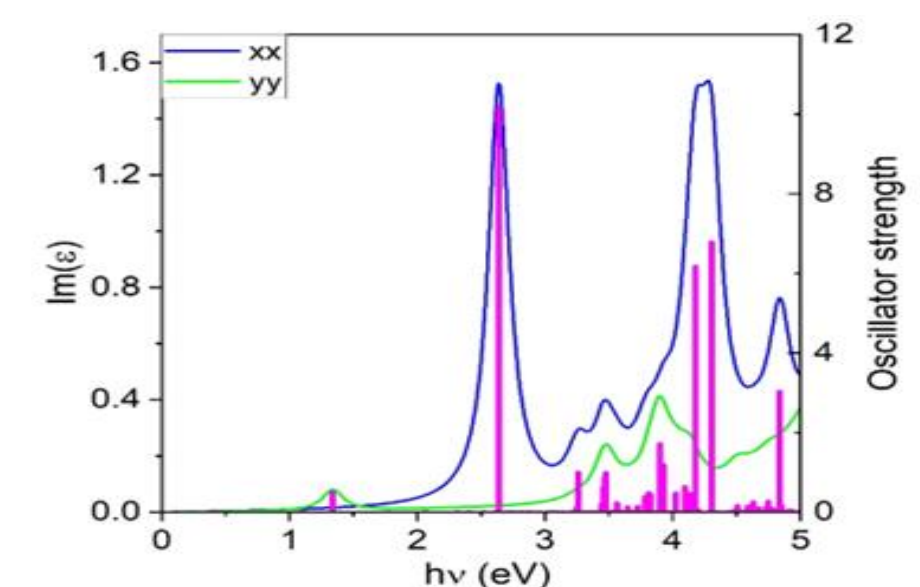


Fig. 4. The frequency-dependent dielectric function (blue and green lines) and the oscillator strength (pink bars) calculated for the singlet states in the V_{BO_N} defect in hBN. The zz-polarization makes a negligible contribution to $\text{Im}(\epsilon)$.

Acknowledgment

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